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Linux-palvelimet - ICI003AS2A-3021

Raportti 4
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Intro

The course server had already been shut down, and I did this module at the end of the course. I used my own server instead. Its name is zunaserver, I log in as pitzuna, and from the Debian virtual machine the address is 192.168.0.100. Before I changed the SSH settings, Apache, or the firewall, I saved a recovery copy of the system so I can put the server back when the course is over.

Login

I connected from the Debian virtual machine and ran apt-get upgrade. Nothing was upgraded since I've ran this command myself a few hours ago. hostname printed zunaserver and whoami printed pitzuna.

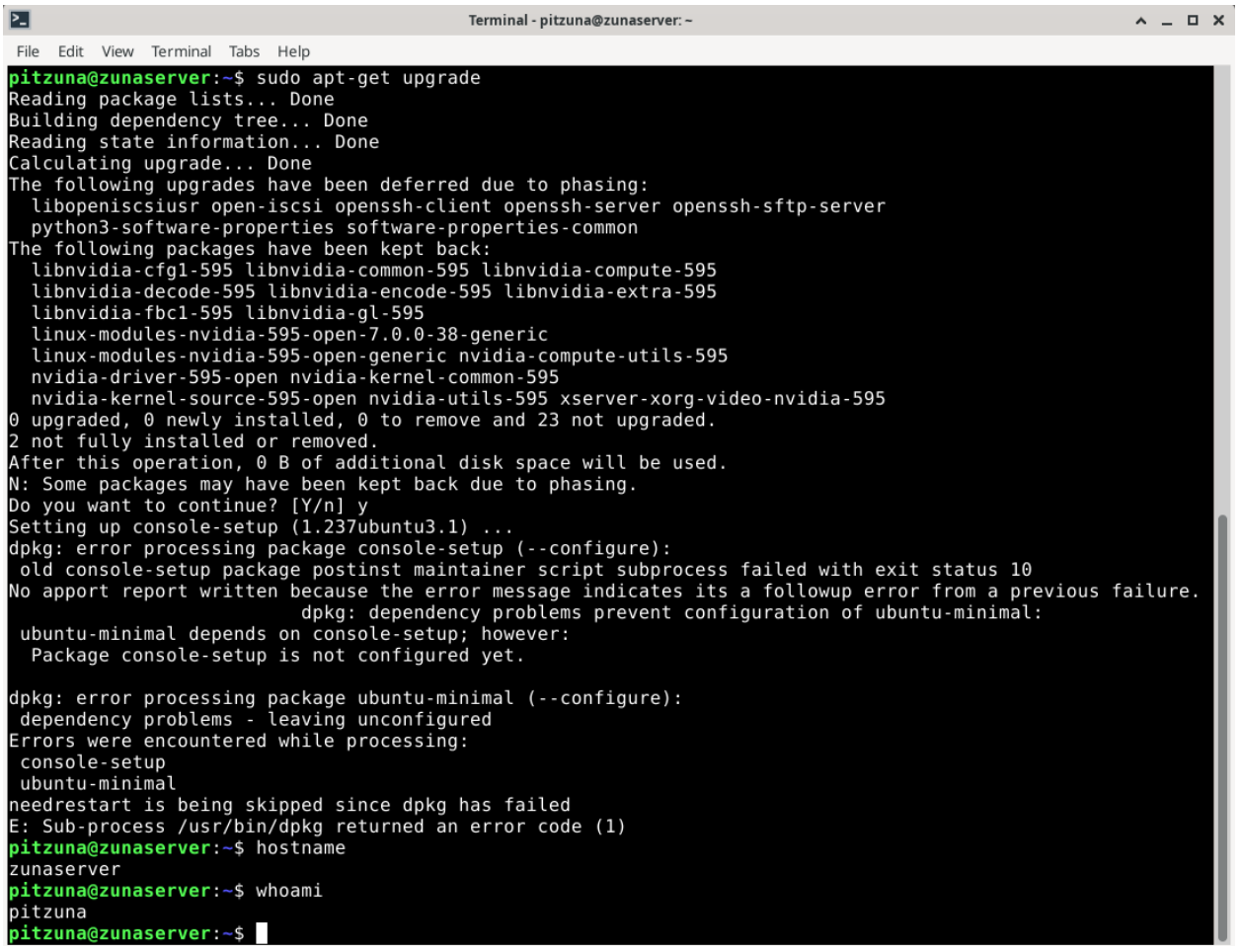


Image 1. zunaserver.

SSH

The SSH service was already running. The unit is marked disabled because ssh.socket starts it. The server was listening on port 22.

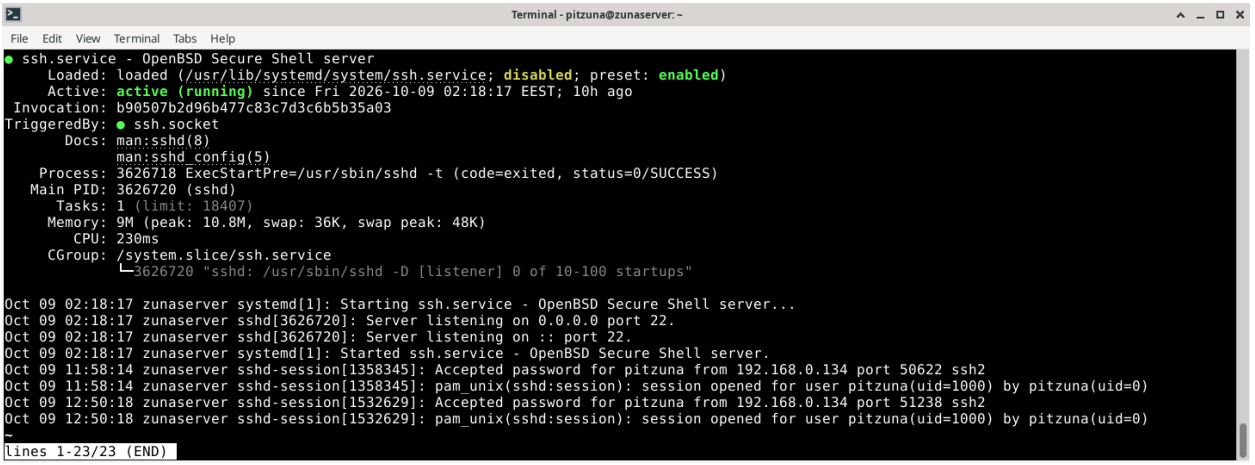


Image 2. SSH is running.

On the Debian virtual machine I created an ed25519 key. The private key stayed on that machine.

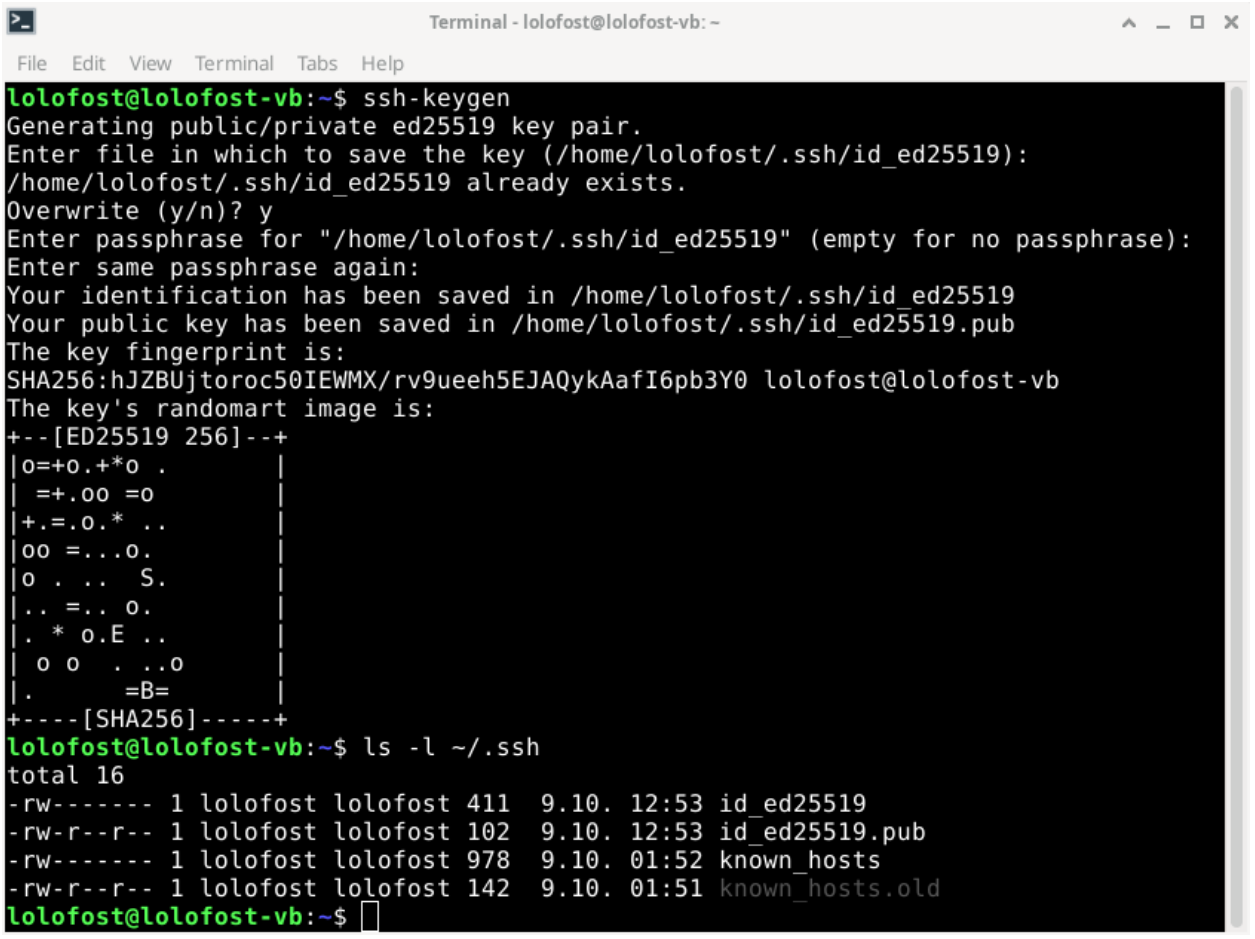


Image 3. The key pair on the Debian virtual machine.

ssh-copy-id asked for the password once and added one public key. The next ssh did not ask for a password. whoami and hostname showed that I was on zunaserver as pitzuna.

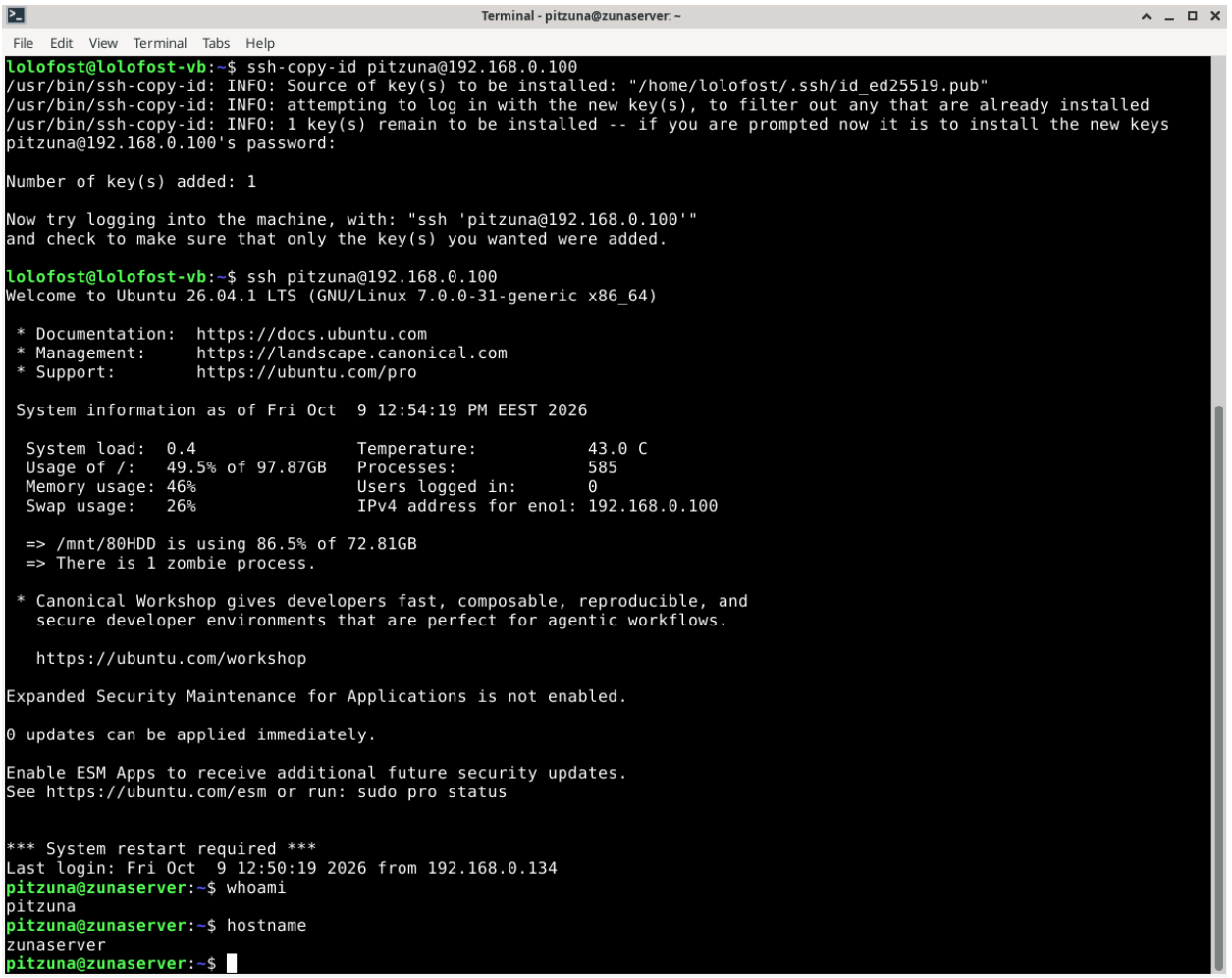


Image 4. Login with the key.

I turned password login off in /etc/ssh/sshd_config.d/50-cloud-init.conf. A matching line in the main config file did not change anything, because Ubuntu reads that directory first. After I restarted ssh, the effective settings were passwordauthentication no, kbdinteractiveauthentication no, and permitrootlogin no. A login that could use only a password was refused. The key login still opened the session.

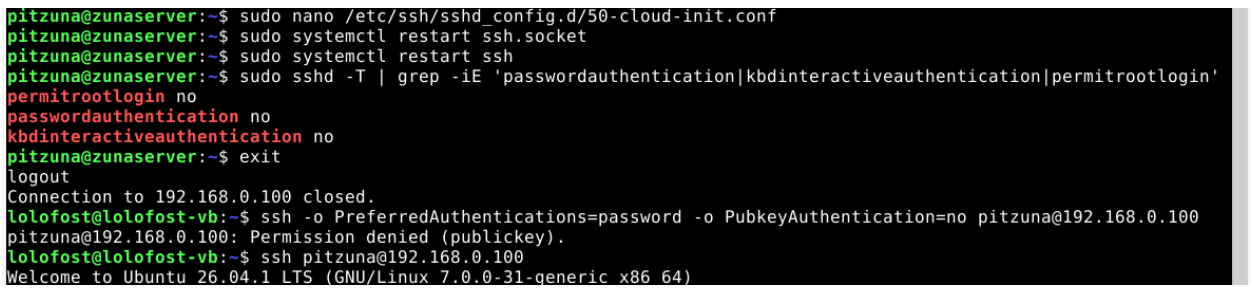


Image 5. Password login refused.

Apache

Port 80 on this server already belongs to Nginx Proxy Manager, so Apache does not use it.

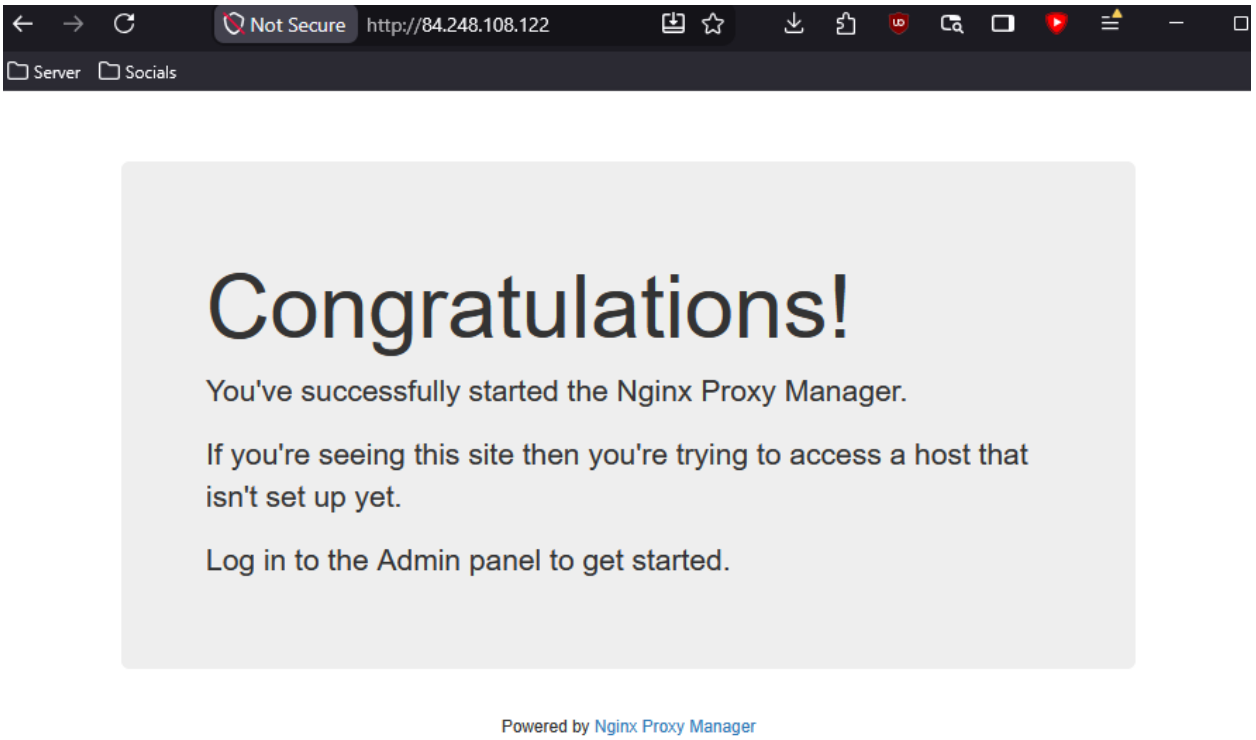


Image 6. Nginx Proxy Manager on port 80.

I set Apache to listen on port 3456 and replaced the default page with the sentence "This is my test default page". In the browser, http://192.168.0.100:3456 showed that sentence. I did not publish port 3456 on the router, so the page opens from the home network. I also created /home/pitzuna/public-sites for the shared-directory exercise.

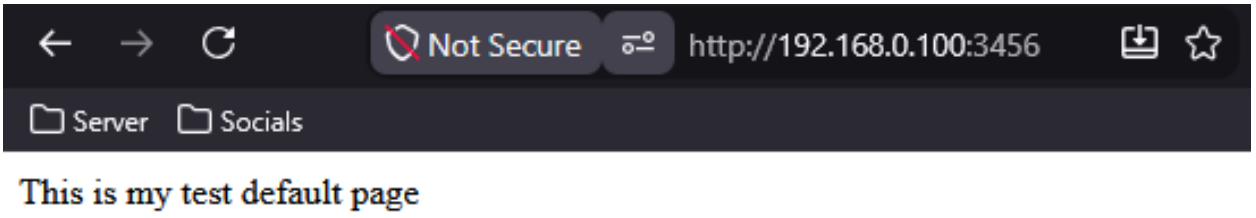


Image 7. The test page in the browser.

curl to 127.0.0.1 on port 3456 printed the same sentence.

```
pitzuna@zunaserver:~$ curl http://127.0.0.1:3456
This is my test default page
pitzuna@zunaserver:~$
```

Image 8. The same page from curl.

Firewall

ufw was already active. Incoming traffic is denied unless a rule allows it. I allowed ports 22, 80, 443, and 3456. The Samba rules for 192.168.0.0/24 were added outside of the course, before.

```
Terminal - pitzuna@zunaserver: ~
File  Edit  View  Terminal  Tabs  Help
pitzuna@zunaserver:~$ sudo ufw status verbose
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), deny (routed)
New profiles: skip

To                         Action      From
--                         ------      ----
137,138/udp (Samba)        ALLOW IN    192.168.0.0/24
139,445/tcp (Samba)        ALLOW IN    192.168.0.0/24
22/tcp                     ALLOW IN    Anywhere
80/tcp                     ALLOW IN    Anywhere
443/tcp                    ALLOW IN    Anywhere
3456/tcp                   ALLOW IN    Anywhere
22/tcp (v6)                ALLOW IN    Anywhere (v6)
80/tcp (v6)                ALLOW IN    Anywhere (v6)
443/tcp (v6)               ALLOW IN    Anywhere (v6)
3456/tcp (v6)              ALLOW IN    Anywhere (v6)

pitzuna@zunaserver:~$
```

Image 9. The firewall.

Addresses

ip addr filled the terminal with Docker bridges, so that list is not in a picture. The address on eno1 is 192.168.0.100/24. The default route goes through 192.168.0.1 on eno1. curl ifconfig.me printed 84.248.108.122. That public address is not on eno1. It is the address of the router.

Packets

I ran ngrep on eno1 for port 3456 and requested the page from the Debian virtual machine. The request was GET / HTTP/1.1 for host 192.168.0.100:3456. The reply was HTTP/1.1 200 OK from Apache/2.4.46, and the page sentence was in the packet. HTTP

does not encrypt the page, so the text is visible. -d chooses the interface, -W byline prints the payload line by line, the empty quotes match every packet, and port 3456 leaves the other traffic out.

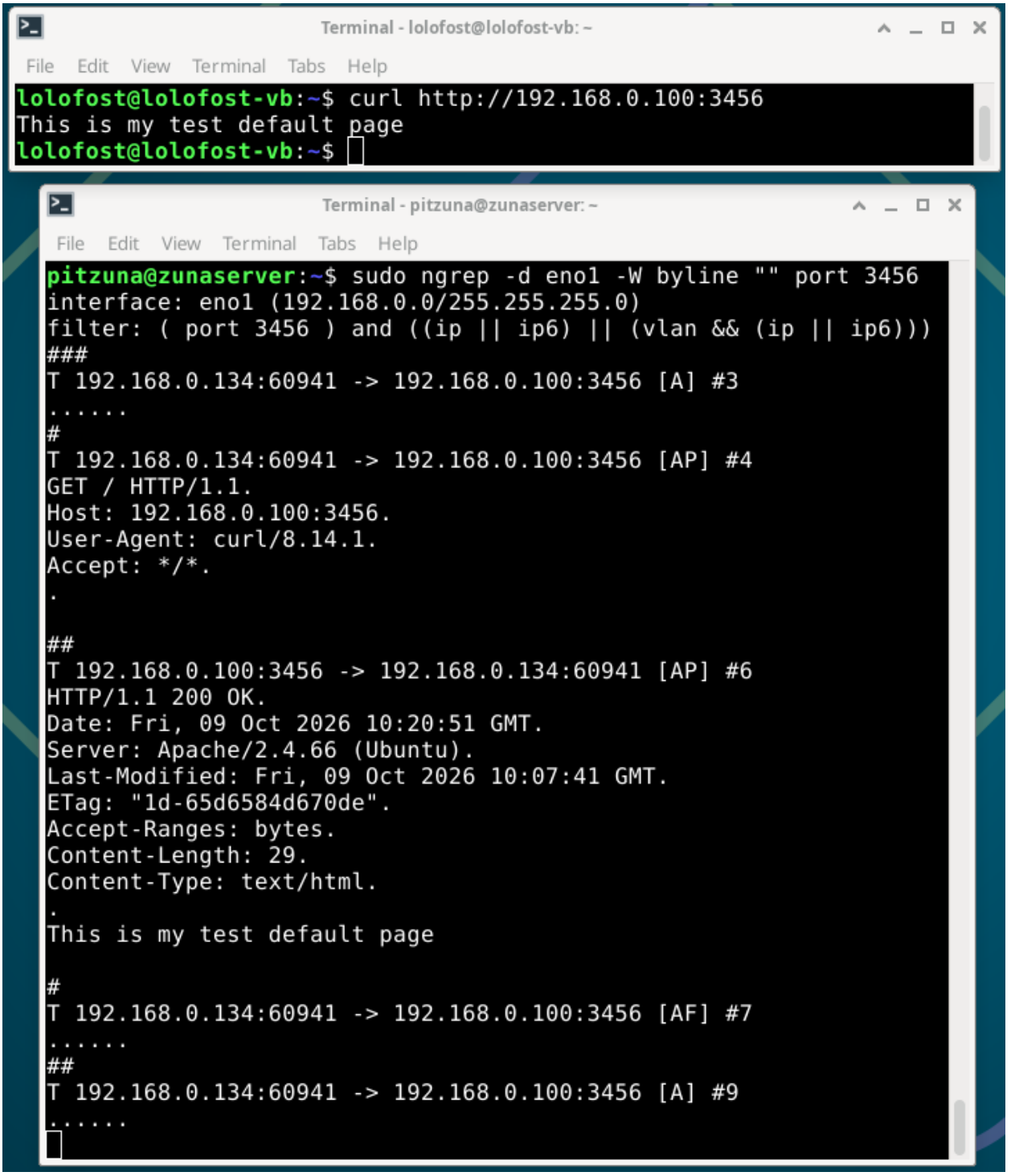


Image 10. The HTTP capture.

On port 22 the capture showed a connection from 192.168.0.100 to 192.168.0.134. The payload was not readable text. SSH encrypts the session, so the commands do not appear.

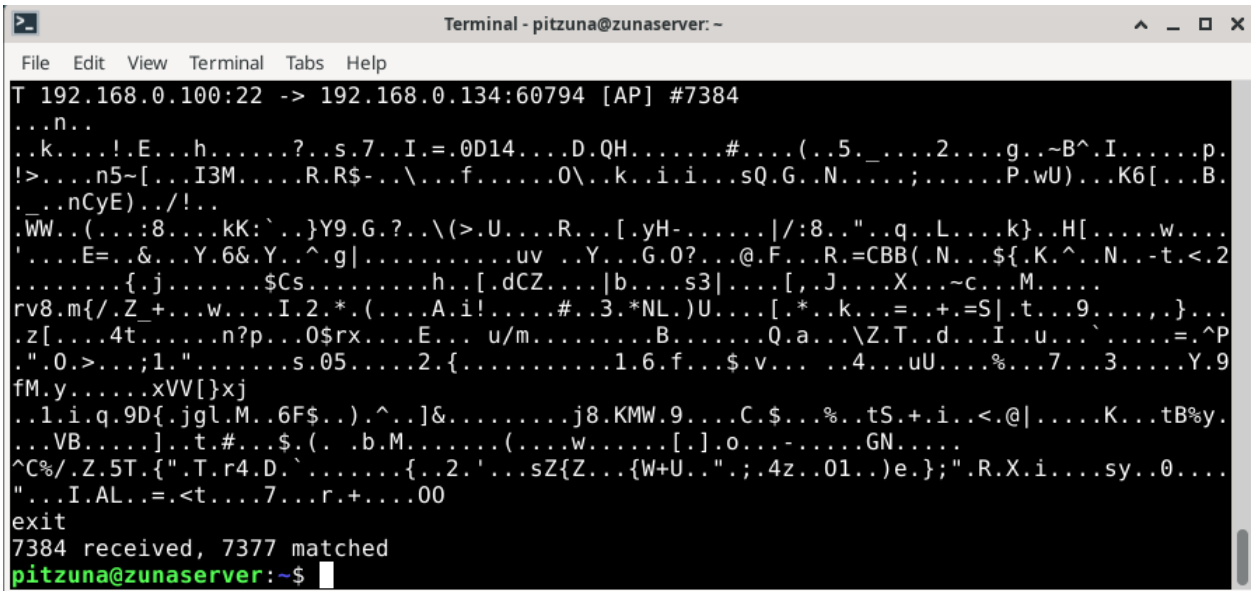


Image 11. The SSH capture.

ping sent three requests to 8.8.8.8 and received three replies, with no loss. The times were about 2 ms. ngrep showed the same requests and replies as raw characters. tcpdump named them ICMP echo request and ICMP echo reply. ICMP is what ping uses to check that a packet can reach an address and come back. tcpdump labels the packet, and ngrep tries to print the contents. Those contents are not a web page.

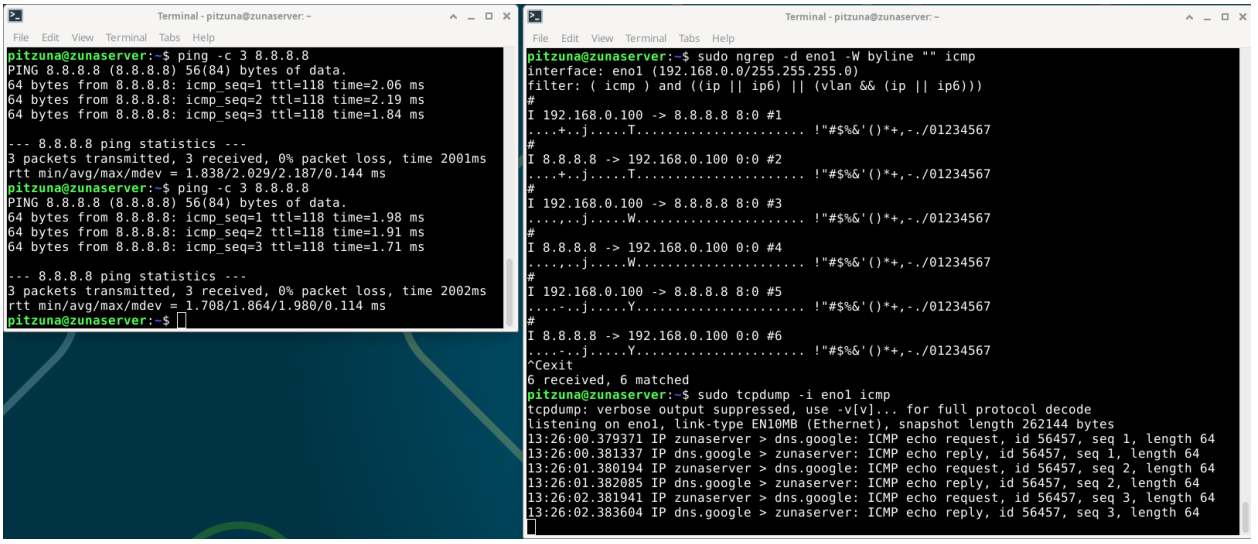


Image 12. Ping with ngrep and tcpdump.

A shared directory

edituser has no sudo. Both accounts are in the group webteam. public-sites is drwxrwsr-x, owned by pitzuna with the group webteam. The s means a new file in that directory takes the group webteam. edituser created from-edituser.txt and added a line to from-pitzuna.txt. From the Debian virtual machine I uploaded from-sftp.txt with SFTP, using the key. All three files are in the group webteam.

```
pitzuna@zunaserver:~$ ls -ld /home/pitzuna/public-sites
drwxrwsr-x 2 pitzuna webteam 4096 Oct  9 13:33 /home/pitzuna/public-sites
pitzuna@zunaserver:~$ ls -l /home/pitzuna/public-sites
total 12
-rw-rw-r-- 1 edituser webteam 14 Oct  9 13:30 from-edituser.txt
-rw-rw-r-- 1 pitzuna  webteam 33 Oct  9 13:30 from-pitzuna.txt
-rw-rw-r-- 1 pitzuna  webteam 10 Oct  9 13:33 from-sftp.txt
pitzuna@zunaserver:~$
```

Image 13. The shared directory.

Notes

The course server was already off, so the work was done using my own alreadyconfigured server, unfortunately. It already had SSH, a firewall, and a web proxy on port 80. I still tested Apache, on port 3456 from the home network.